

Cloud Computing Course

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1.1

```
54.161.15.213
Terminal Sessions View Xserver Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
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Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
Verifying : containerd-2.2.4-1.amzn2023.0.1.x86_64 2/11
Verifying : docker-25.0.16-1.amzn2023.0.2.x86_64 3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 4/11
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/11
Verifying : libcgrouper-3.0-1.amzn2023.0.1.x86_64 6/11
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 7/11
Verifying : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64 8/11
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 9/11
Verifying : pigz-2.5-1.amzn2023.0.4.x86_64 10/11
Verifying : runc-1.3.5-1.amzn2023.0.2.x86_64 11/11

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch containerd-2.2.4-1.amzn2023.0.1.x86_64 docker-25.0.16-1.amzn2023.0.2.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 libcgrouper-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64 libnftnl-1.2.2-2.amzn2023.0.2.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64 runc-1.3.5-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-22-93 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-22-93 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-22-93 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-22-93 ~]$
```

1.2

```
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Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-22-93 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:9649ff522e70807ab6384a5c0485a79b9c7c08ca79ba08623edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[ec2-user@ip-172-31-22-93 ~]$
```

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1.3

```

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Terminal Sessions View Xserver Tools Games Settings Macros Help
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Quick connect... 3. 54.161.15.213

2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
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For more examples and ideas, visit:
https://docs.docker.com/get-started/

[ec2-user@ip-172-31-22-93 ~]$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
72c03230f136: Pull complete
d4dcde3aeed: Pull complete
c5a7565de4cf: Pull complete
57b3bf43092: Pull complete
80a08b690ad: Pull complete
41c9f6f90940: Pull complete
6376488be516: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-22-93 ~]$

```

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1.4

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to 5.com/nginx.

Thank you for using nginx.

2.1

```

[ec2-user@ip-172-31-22-93 ~]$ sudo usermod -aG docker ec2-user
[ec2-user@ip-172-31-22-93 ~]$ newgrp docker
[ec2-user@ip-172-31-22-93 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
a7e0a35e6584   nginx    "/docker-entrypoint..." 3 minutes ago Up 3 minutes    0.0.0.0:80->80/tcp, :::80->80/tcp    great_edison
[ec2-user@ip-172-31-22-93 ~]$ docker volume create aribah
aribah
[ec2-user@ip-172-31-22-93 ~]$ docker volume ls
DRIVER    VOLUME NAME
local     aribah
[ec2-user@ip-172-31-22-93 ~]$

```

```

[ec2-user@ip-172-31-22-93 ~]$ docker run -it --aribah-container -v aribah:/data ubuntu bash
unknown flag: --aribah-container
See 'docker run --help'.
[ec2-user@ip-172-31-22-93 ~]$ docker run -it --name aribah-container -v aribah:/data ubuntu bash
root@439344be8f4d:/# ls /
bin boot data dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@439344be8f4d:/# echo "Name: Aribah Shirgaonker" > /data/info.txt
root@439344be8f4d:/# echo "Roll No. 24108A0016" >> /data/info.txt
root@439344be8f4d:/# cat /data/info.txt
Name: Aribah Shirgaonker
Roll No. 24108A0016
root@439344be8f4d:/#

```

3.1

```

[ec2-user@ip-172-31-22-93 ~]$ nano Dockerfile
[ec2-user@ip-172-31-22-93 ~]$ cat Dockerfile
FROM alpine

CMD ["echo", "Cloud Computing practical exam"]
[ec2-user@ip-172-31-22-93 ~]$

```

3.2	<pre> [ec2-user@ip-172-31-22-93 ~]\$ docker build -t cloud-exam . [+] Building 0.8s (5/5) FINISHED => [internal] load build definition from Dockerfile => => transferring dockerfile: 156B => [internal] load metadata for docker.io/library/alpine:latest => [internal] load .dockerignore => => transferring context: 2B => [1/1] FROM docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4 => => resolve docker.io/library/alpine:latest@sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4 => => extracting sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63 => => sha256:a2d49ea686c2adfe3c992e47dc3b5e7fa6e6b5055609400dc2acaeb241c829f4 9.22kB / 9.22kB => => sha256:33154315cf4402e697f065e6ec2156e292187e633908ccfed9c66279b6fa956 1.02kB / 1.02kB => => sha256:772bee4540c411d1fec755cb8ead76db4173d38afde7b2018a08f255907bbf23 608B / 608B => => sha256:9b70e313681f44d32991ec943f89228bc91d7431d4a84feafc269a76e3f96a63 3.87MB / 3.87MB => => exporting to image => => exporting layers => => writing image sha256:e5301451a74a00c30607e9e6a360094a520eb49e0689a42646be1ac2f1747c3f => => naming to docker.io/library/cloud-exam [ec2-user@ip-172-31-22-93 ~]\$ docker images REPOSITORY TAG IMAGE ID CREATED SIZE nginx latest 936fef290c8f 2 days ago 161MB cloud-exam latest e5301451a74a 3 days ago 8.45MB ubuntu latest 30ba44506a6d 7 weeks ago 100MB hello-world latest e2ac70e7319a 2 months ago 10.1kB [ec2-user@ip-172-31-22-93 ~]\$ </pre>
3.3	<pre> [ec2-user@ip-172-31-22-93 ~]\$ docker images REPOSITORY TAG IMAGE ID CREATED SIZE nginx latest 936fef290c8f 2 days ago 161MB cloud-exam latest e5301451a74a 3 days ago 8.45MB ubuntu latest 30ba44506a6d 7 weeks ago 100MB hello-world latest e2ac70e7319a 2 months ago 10.1kB [ec2-user@ip-172-31-22-93 ~]\$ docker run cloud-exam Cloud Computing practical exam [ec2-user@ip-172-31-22-93 ~]\$ </pre>
3.4	Difference between a Docker Image and a Docker Container.
	Docker image is the read only template also known as recipe, while container is the running instance of the image. If image is the recipe container is the finished dish.
3.5	Why are Docker Volumes used?
	Volumes are used to store data even if the container is deleted data stays in the volume. This keeps the data safe and also allows it to attach to other containers.